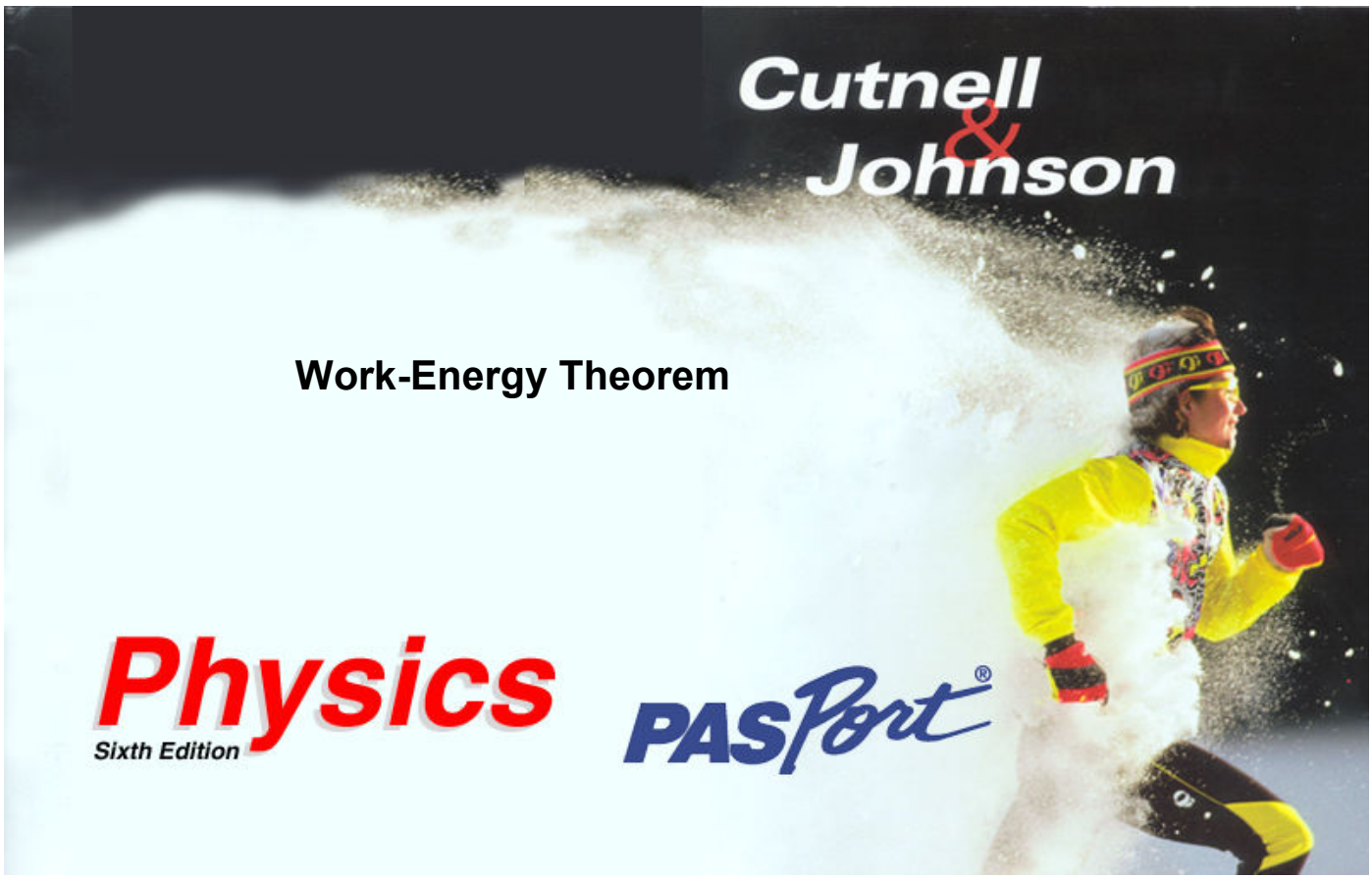

Cutnell
&
Johnson
Work-Energy Theorem
Physics
Sixth Edition
PASPort®

Student Information

§ Enter the appropriate information in the boxes below.

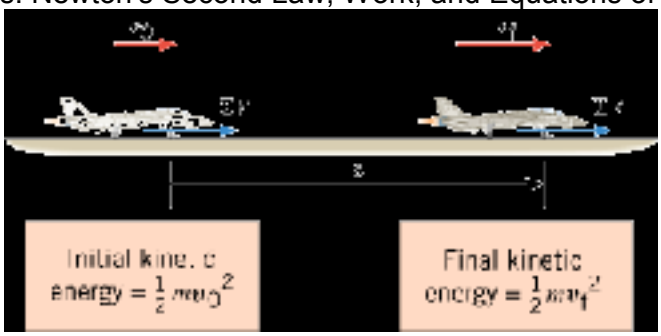
Date:	
Your Name:	
Group:	
Period:	

Introduction *Exploration Goals*

§ The purpose of this exploration is to compare the work done on an object to the change in kinetic energy of the object.

§ Use the Force Sensor to measure the force applied to the cart. Use the Photogate/Pulley System to measure the motion of the cart as it is pulled by the weight of the hanging mass.

§ Most people expect that if you do work, you get something as a result. In physics, when a net force performs work on an object, there is always a result from the effort. The result is a change in the energy of the object. The relationship that relates work to the change in kinetic energy is known as the work-energy theorem. This theorem is obtained by bringing together three basic concepts: Newton's Second Law, Work, and Equations of Kinematics.



Learning Outcomes

You will be able to:

- Qualitatively understand the Work-Energy Theorem.
- Make measurements to determine the amount of work done on an object.
- Use sensors to determine the change in kinetic energy of an object.
- Compare the change in kinetic energy of an object to the area under a graph of force versus distance.

Hypothesis

§ How does the amount of work done on an object to make it move in the horizontal direction compare to the amount of kinetic energy gained by the object?

Background

§ For an object with mass **m** that experiences a net force **Fnet** over a distance **d** that is parallel to the net force, the work done is shown by Equation 6.1.

$$W = F_{net} d$$

Equation 6.1

§ If the work changes the object's vertical position, the object's gravitational potential energy changes. However, if the work changes only the object's speed, the object's kinetic energy changes as shown in Equation 6.2 where **W** is the work, **vf** is the final speed of the object and **vi** is the initial speed of the object.

» For more information see Cutnell & Johnson, Physics, 6th ed., Volume One, Chapter 6, Section 6.2.

$$W = \Delta KE = KE_f - KE_i = \frac{1}{2}mv_f^2 - \frac{1}{2}mv_i^2$$

Equation 6.2

Materials

- 2 PASPORT USB Links
- PASPORT Force Sensor
- PASPORT Photogate Port
- Photogate/Pulley System
- PAScar with Mass Bar
- End Stop with Magnets
- String
- Mass and Hanger Set
- 1.2 m Dynamics Track
- Balance
- Universal Table Clamp

Setup

Equipment Setup

Safety Reminder: Follow the directions for using the equipment.

§ Use the thumbscrew that comes with the Force Sensor to mount the sensor onto the accessory tray of the cart.

§ Add a 250-g mass to the cart.

§ Measure the total mass of the cart, Force Sensor, and 250-g mass. Convert the mass to kilograms and record the value.

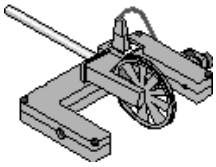
§ Place the Dynamics Track on a horizontal surface. Level the Dynamics Track by placing the Dynamics Cart on the Dynamics Track. If the cart rolls one way or the other, use the Adjustable Feet at one end of the Dynamics Track to raise or lower that end until the Dynamics Track is level and the cart does not roll one way or the other.

Mass of cart, Force Sensor, and 250-g mass:

Note: It is very important that the track is level to get the best results.

Setup

Equipment Setup



§ Use the Pulley Mounting Rod to attach the Pulley to the tab on the Photogate.

§ Attach the Universal Table Clamp on the end of the track. Mount the Photogate/Pulley System's rod in the clamp so that the top edge of the pulley is approximately the same height as the hook on the Force Sensor that is mounted on the cart.

§ Place the end stop with magnets about 1 cm in front of the smart pulley. Arrange the cart so that the magnets on the cart and the magnets on the end stop face one another.

§ Use a piece of string that is about 10 centimeters longer than the distance from the top of the pulley to the floor. Connect one end of a string to the Force Sensor's hook. Place the string in the Pulley's groove.

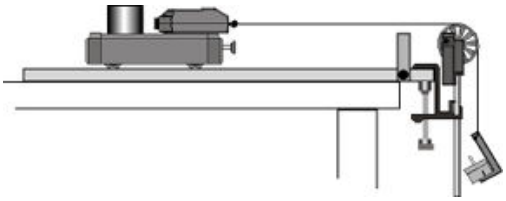
Setup

Equipment Setup

§ Put a known mass (e.g., 50-g) on the mass hanger. Weigh the hanger and the mass. Convert the mass to kilograms and record the value.

§ Press the **Zero** button on the Force Sensor to zero the sensor.

Hanger and mass:



Note: The diagram is not to scale.

§ Attach the hanger with the known mass to the end of the string in such a way that the bottom of the hanger is just above the floor when the cart is against the end-stop.

Process *Enter the Total Mass into the KE Calculation*

§ Find the total mass of the system (cart, sensor, 250-g mass, hanger, etc.).

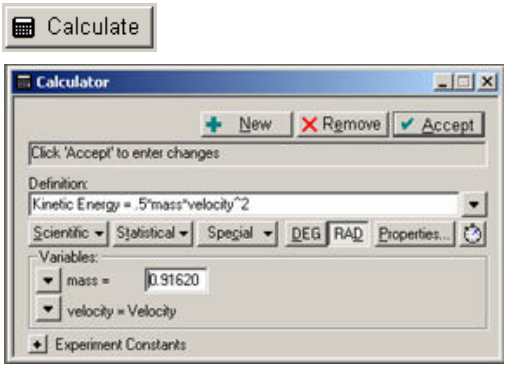
» The calculation for Kinetic Energy uses the total mass of the system.

Mass of cart, Force Sensor, and 250-g mass: _____ Hanger and mass: _____

§ Add the masses together to find the total mass of the system. Record the total mass.

Total mass: _____

§ Click the **Calculate** button to open the Calculator window. Enter the value for the total **mass** and then click **Accept**. Close the window after you enter the mass.



Process *Record Data*

Reminder: Don't let the cart crash into the end stop.

§ Pull the cart away from the Photogate so the mass on the end of the string is just below the pulley.

§ Hold the Force Sensor cable so the cart can move freely. Click **Start**. Release the cart so that it can move toward the Photogate.

§ Click **Stop** just before the cart hits the end stop.

» The Table shows the Velocity and Kinetic Energy.

Maximum Kinetic Energy:

• Kinetic Energy vs Velocity
No Data

	Velocity (m/s)	Kinetic Energy
Maximum		

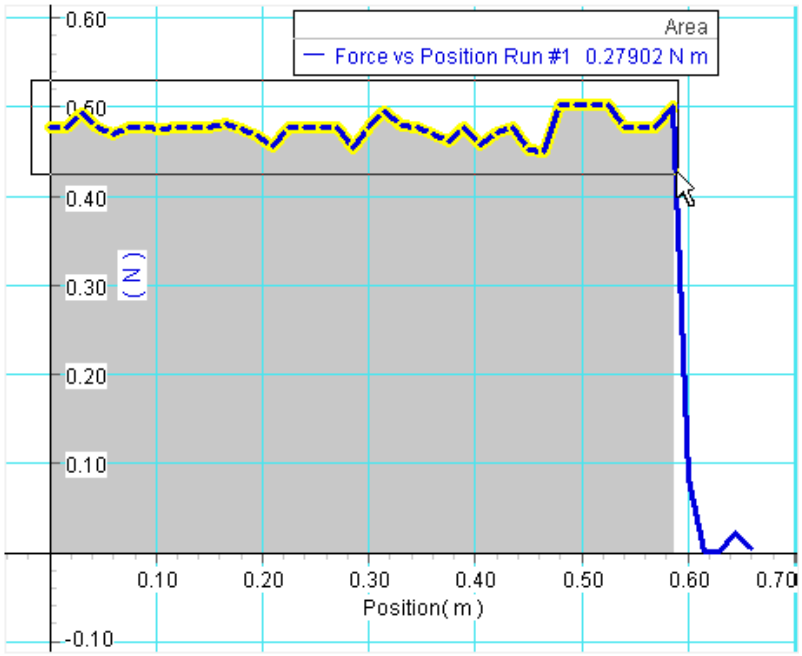
Analyze *How to Find the Area Under the Graph*

§ Find the area under the curve for the Force vs. Position graph.

§ Click the **Scale to Fit** button to rescale the graph.

§ Highlight the flat part of the graph. (See the example.)

» The value for the area under the curve will appear in the Legend box. In this example, the area is 0.27902 N m.



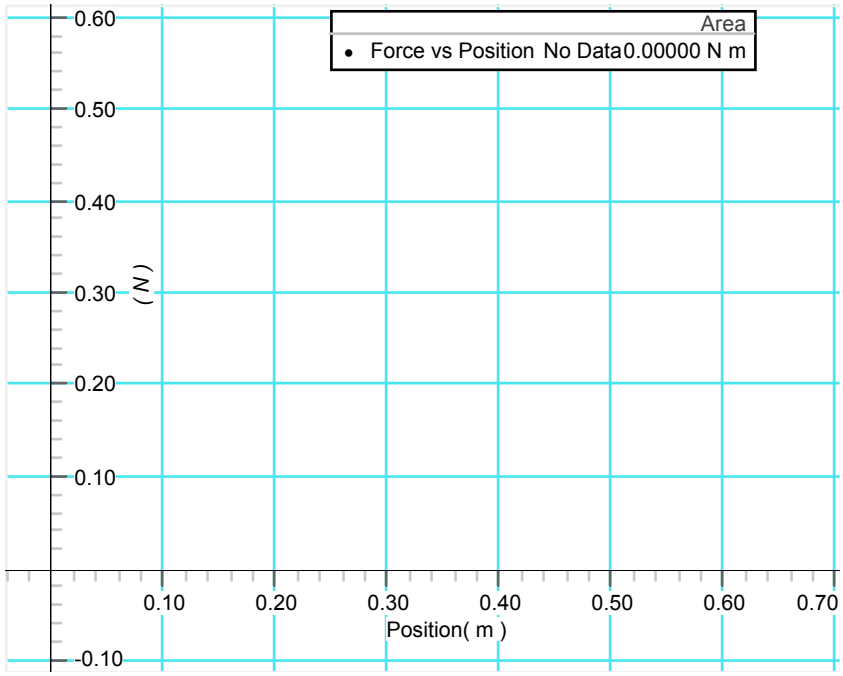
Analyze

Find the Area Under the Graph

§ Use the procedure described previously to find the area under the graph.

§ Record the area below.

Area:



Observations

§ Why is the graph of the force versus position not uniform? Explain.

Analyze *Comparison and Verification*

§ How does the maximum kinetic energy compare to the area under the curve? Explain.

Maximum Kinetic Energy:

Area:

§ Compute the percent difference between the Maximum Kinetic Energy and the Area under the force-position graph. Use the Area under the graph as the *theoretical* value.

% difference:

$$\% \text{ difference} = \left| \frac{\text{measured-theoretical}}{\text{theoretical}} \right| \times 100$$

Synthesize *Conclusions and Error Analysis*

§ What is the relationship between kinetic energy and work? Explain.

§ How do you know when work is being done?

§ What were the sources of error in this experiment?

§ How does your hypothesis compare to your results? Hypothesis:

Synthesize *Applications*

» The rocket boosters on the sides of the Space Shuttle Endeavor supply a constant force of 14.4 million newtons each.

» They are used to lift the shuttle from the ground to an altitude of roughly 45.7 kilometers.

§ How much work do these boosters do in one launch?

Note: This is about as much energy as the average American family will use in their household over the next 40 years!

Synthesize *Extension Problem*

§ In Cutnell & Johnson, Physics, 6th ed., Volume One, Chapter 6, Problem 12, page 175.

A fighter jet is launched from an aircraft carrier with the aid of its own engines and a steam-powered catapult. The thrust of its engines is 2.3 x 10^5 N. In being launched from rest it moves a distance of 87 m and has a kinetic energy of 4.5 x 10^7 J at lift-off. What is the work done on the jet by the catapult?

Notes